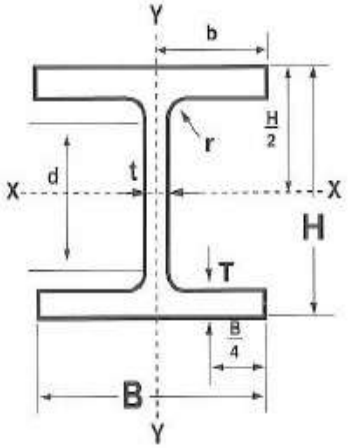
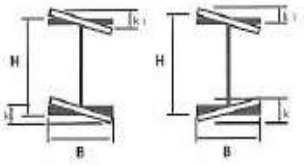
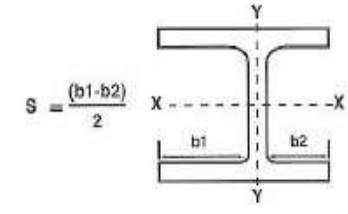
4(iv) Dimension & Tolerances ASTM A6 (2003)

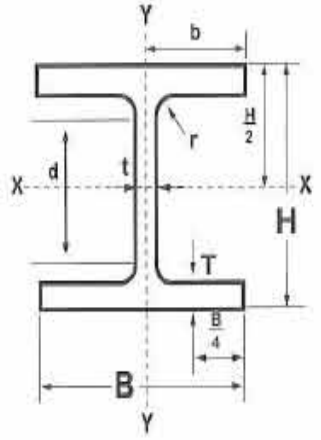
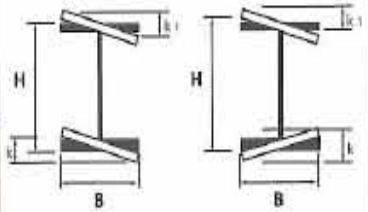
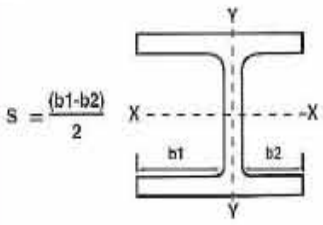
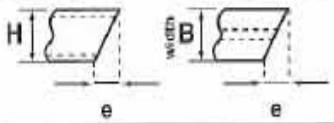
ASTM A 6 : 2003

H-Beams

Dimension (mm)			Tolerance (mm)	Remarks
Height (H)	H ≤ 310		- 3.0 + 4.0	
	H > 310		- 3.0 + 4.0	
Width (B)	B ≤ 310		- 5.0 + 6.0	
	B > 310		- 5.0 + 6.0	
Length (L)	H ≤ 610	L ≤ 9 m	+ 10 - 10	
		L > 9 m	10+(No. of meter of L-9) - 10	
	H > 610	L ≤ 9 m	+ 13 - 13	
		L > 9 m	13+(No. of meter of L-9) - 13	
Out-of-square (k, k_1)	H ≤ 310		≤ 3.0	
	H > 310		≤ 4.0	
Bend	L ≤ 14 m	1 mm x no. of meters of L, Max 10 mm		To be applied to bend such as sweep and camber
	L > 14 m	10 mm + [1mm x (no. of meters of L-14m)]		
Web-off-center (S)	H ≤ 310		≤ 5.0	
	H > 310		≤ 5.0	
Mass (kg / m)	± 2.5%			

Standard Specifications

4 (vi) Dimension & Tolerances JIS A 5526 : 1988

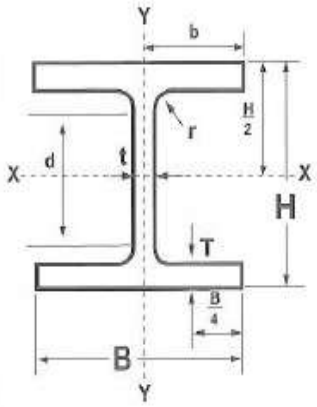
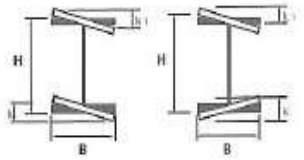
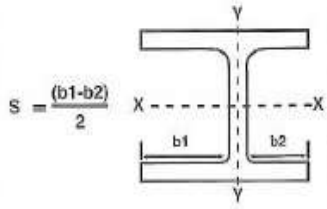
JIS A 5526				
H-Bearing Piles				
Dimension		Dimension	Tolerance	Remark
Height (H*)		H < 400 H ≥ 400	± 3.0 ± 4.0	
Width (B*)		-	± 3.0	
Thickness (t, T)	Web, t	t ≤ 13	+Not specified - 0.8	
		t > 13	+Not specified - 6%	
	Flange,T	T ≤ 13	+Not specified - 0.8	
		T > 13	+Not specified - 6%	
Length (L)		-	+Not specified 0	
Out-of-square (k, k_1)		H ≤ 300 H > 300	1.2% of B 1.5% of B	
Bend		H ≤ 300 H > 300	0.20% of L 0.10% of L	To be applied to bend such as sweep and camber
Web-off-center (S)		H ≤ 300	± 3.0	
		H > 300	± 4.5	
Concavity of web (W)		-	Not specified	
Ends-Out-of square (e)		-	1.6% of B or H	
Mass (kg / m)		-	Not specified	1. Thicker nominal values shall be applied. 2. To be applied to one lot of the same size (1t or over), provided that, when the number of pieces corresponding to 1t does no amount to 10 pieces, is shall be applied to each lot of 10 or more pieces.

4 (iii) Rolling Tolerances - European Standard BS EN 10034:1993

Dimension & Tolerances

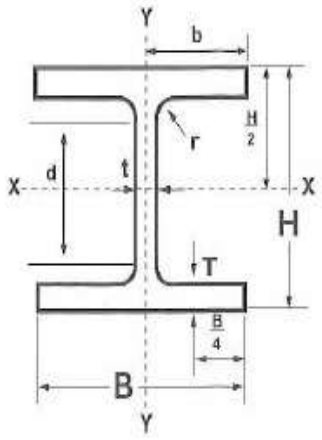
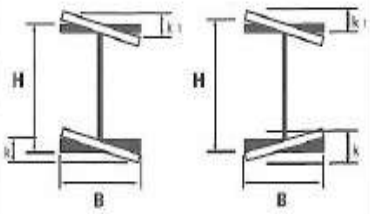
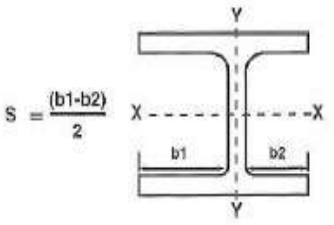
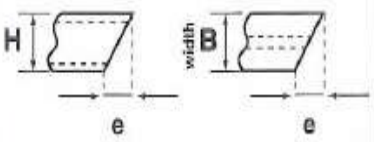
BS EN 10034 : 1993

Hot Rolled Universal Beams and Columns

Dimension (mm)				Tolerance (mm)	Remarks
Height (H)		Up to and including 180	H ≤ 180	+ 3.0, - 2.0	
		Greater than 180 up to and including 400	180 < H ≤ 400	+ 4.0, - 2.0	
		Greater than 400 up to and including 700	400 < H ≤ 700	+ 5.0, - 3.0	
		Greater than 700	H > 700	+ 5.0, - 5.0	
Width (B)		Up to and including 110	B ≤ 110	+ 4.0, - 1.0	
		Greater than 110 up to and including 210	110 < B ≤ 210	+ 4.0, - 2.0	
		Greater than 210 up to and including 325	210 < B ≤ 325	+ 4.0, - 4.0	
		Greater than 325	B > 325	+ 6.0, - 5.0	
Thickness (t, T)	Web, t	Less than 7	t < 7	± 0.7	
		7 up to but excluding 10	7 ≤ t < 10	± 1.0	
		10 up to but excluding 20	10 ≤ t < 20	± 1.5	
		20 up to but excluding 40	20 ≤ t < 40	± 2.0	
		40 up to but excluding 60	40 ≤ t < 60	± 2.5	
		60 and above	t ≥ 60	± 3.0	
	Flange,T	Less than 6.5	T < 6.5	+ 1.5, - 0.5	
		6.5 up to but excluding 10	6.5 ≤ T < 10	+ 2.0, - 1.0	
		10 up to but excluding 20	10 ≤ T < 20	+ 2.5, - 1.5	
		20 up to but excluding 30	20 ≤ T < 30	+ 2.5, - 2.0	
		30 up to but excluding 40	30 ≤ T < 40	+ 2.5, - 2.5	
		40 up to but excluding 60	40 ≤ T < 60	+ 3.0, - 3.0	
		60 and above	T ≥ 60	+ 4.0, - 4.0	
Length (L)		-	-	+ 50 - 50	
Out-of-square (k, k_1)		Up to and including 110	B ≤ 110	1.5	
		Greater than 110	B > 110	2% of B (Max 6.5)	
Bend		Greater than 80 up to and including 180	80 < H ≤ 180	0.30% of L	To be applied to bend such as sweep and camber
		Greater than 180 up to and including 360	180 < H ≤ 360	0.15% of L	
		Greater than 360	H > 360	0.10% of L	
Web-off-center (S)		Up to and including 110	B ≤ 110	2.5	
		Greater than 110 up to and including 325	110 < B ≤ 325	3.5	
		Greater than 325	B > 325	5.0	
Mass (kg / m)		-	-	± 4%	The calculated mass shall be determined using a density of 7.85 kg / cm^3

Standard Specifications

4(v) Dimension & Tolerances JIS G 3192 : 1990/1994

JIS G 3192				
H-Beams				
Dimension			Tolerance	Remark
Height (H*)		H < 400 400 ≤ H < 600 H ≥ 600	± 2.0 ± 3.0 ± 4.0	
Width (B*)		B < 100 100 ≤ B < 200 B ≥ 200	± 2.0 ± 2.5 ± 3.0	
Thickness (t, T)	Web, t	t < 16 16 ≤ t < 25 25 ≤ t < 40 t ≥ 40	± 0.7 ± 1.0 ± 1.5 ± 2.0	
	Flange,T	T < 16 16 ≤ T < 25 25 ≤ T < 40 T ≥ 40	± 1.0 ± 1.5 ± 1.7 ± 2.0	
Length (L)		L ≥ 7m	+ 40 0	
		L > 7m	40+(No. of meter of L-7)x5 0	
Out-of-square (k,k_1)		H ≤ 300 H > 300	1.0% of B, Max. 1.5mm 1.2% of B, Max. 1.5mm	
Bend		H ≤ 300 H > 300	0.15% of L 0.10% of L	To be applied to bend such as sweep and camber
Web-off-center (S)		H ≤ 300, B ≤ 200 H > 300, B > 200	± 2.5 ± 3.5	
Concavity of web (W)		H < 400 400 ≤ H < 600 H ≥ 600	2.0 2.5 3.0	
Ends-Out-of square (e)		All Section	1.6% of B or H Max 3 mm	
Mass (kg / m)		t < 10	± 5%	1. Thicker nominal values shall be applied. 2. To be applied to one lot of the same size (1t or over), provided that, when the number of pieces corresponding to 1t does no amount to 10 pieces, is shall be applied to each lot of 10 or more pieces.
		t ≥ 10	± 4%	

Standard Specifications

4 (ii) Chemical Composition for Structural Steel and Sheet Pile

Classifications		Chemical Compositions (ladle analsis), %									
		C (max.)	Si (max.)	Mn (max.)	P (max.)	S (max.)	CE (max.)	SW (max.)	NB (max.)	V (max.)	N (max.)
JIS G 3101 : 2004	SS400	-	-	-	0.050	0.050	-	-	-	-	-
	SS490	-	-	-	0.050	0.050	-	-	-	-	-
	SS540	0.30	-	1.6	0.040	0.040	-	-	-	-	-
JIS G 3106 : 2004	SM400 A	0.23	-	2.50 x C min.	0.035	0.035	-	-	-	-	-
	SM400 B	0.20	0.35	0.60 - 1.40	0.035	0.035	-	-	-	-	-
	SM490 A	0.20	0.55	1.60	0.035	0.035	-	-	-	-	-
	SM490 B	0.18	0.55	1.60	0.035	0.035	-	-	-	-	-
	SM490 YA	0.20	0.55	1.60	0.035	0.035	-	-	-	-	-
	SM490 YB	0.20	0.55	1.60	0.035	0.035	-	-	-	-	-
	SM520 B	0.20	0.55	1.60	0.035	0.035	-	-	-	-	-
	SM520 C	0.20	0.55	1.60	0.035	0.035	-	-	-	-	-
JIS G 3136 : 2005	SN400 A	0.24	-	-	0.050	0.050	-	-	-	-	-
	SN400 B	0.20	0.35	0.60 - 1.40	0.030	0.015	0.36	0.26	-	-	-
	SN490 B	0.18	0.55	1.60	0.030	0.015	0.44	0.29	-	-	-
BS 4360 : 1990	43A	0.25	0.50	1.60	0.050	0.050	-	-	-	-	-
	43B	0.21	0.50	1.50	0.050	0.050	-	-	-	-	-
	43C	0.18	0.50	1.50	0.050	0.050	-	-	-	-	-
	43D	0.18	0.50	1.50	0.050	0.050	-	-	-	-	-
	43DD	0.18	0.50	1.50	0.050	0.050	-	-	-	-	-
	50A	0.23	0.50	1.60	0.050	0.050	-	-	0.003-0.100	0.003-0.100	-
	50B	0.20	0.50	1.5	0.050	0.050	-	-	0.003-0.100	0.003-0.100	-
	50C	0.20	0.50	1.5	0.050	0.050	-	-	0.003-0.100	0.003-0.100	-
	50D	0.18	0.50	1.5	0.040	0.040	-	-	0.003-0.100	0.003-0.100	-
BS EN 10025 : 1993/2004	S275JR	0.21	-	1.50	0.035	0.035	-	-	-	-	0.009
	S275JO	0.18	-	1.50	0.030	0.030	-	-	-	-	0.009
	S275J2	0.18	-	1.50	0.025	0.025	-	-	-	-	-
	S275J2G3	0.18	-	1.50	0.035	0.035	-	-	-	-	-
	S355JR	0.24	0.55	1.60	0.035	0.035	-	-	-	-	0.009
	S355JO	0.20	0.55	1.60	0.030	0.030	-	-	-	-	0.009
	S355J2	0.20	0.55	1.60	0.035	0.035	-	-	-	-	-
	S355J2G3	0.20	0.55	1.60	0.035	0.035	-	-	-	-	-
ASTM : 2003	A36	0.26	0.40	-	0.040	0.050	-	-	-	-	-
	A 572 Gr.42	0.21	0.40	1.35	0.040	0.050	-	-	-	0.010-0.150	-
	A 572 Gr.50	0.23	0.40	1.35	0.040	0.050	-	-	-	0.010-0.150	-
	A 992	0.23	0.10-0.40	0.50 - 1.50	0.035	0.450	0.45	-	-	0.11	0.012
DIN 17100	St 33	-	-	-	-	-	-	-	0.05	-	-
	St 37-2	0.17	-	-	0.050	0.050	-	-	-	-	0.009
	St 44-2	0.21	-	-	0.050	0.050	-	-	-	-	0.009
	St 50-2	-	-	-	0.050	0.050	-	-	-	-	0.009
	St 52-3	0.2	-	-	0.040	0.040	-	-	-	-	-
AS/NZS 3679.1 : 1996	250	0.25	0.40	-	0.040	0.040	0.43	-	-	-	-
	250L0	0.20	0.40	1.50	0.040	0.040	0.42	-	-	-	-
	350	0.22	0.50	1.60	0.040	0.040	0.45	-	-	-	-
	350L0	0.22	0.50	1.60	0.040	0.040	0.45	-	-	-	-
	300	0.22	0.50	1.60	0.040	0.040	0.44	-	-	-	-
	300L0	0.22	0.50	1.60	0.040	0.040	0.44	-	-	-	-
JIS A 5528 : 2000	SY295	-	-	-	0.040	0.040	-	-	-	-	-
	SY390	-	-	-	0.040	0.040	-	-	-	-	-

Standard Specifications

The Standard Specifications used for production of universal beams and colums in this region are listed in this table.

4 (i) Mechanical Properties and Dimension & Tolerance for Structural Steel and Sheet Pile

Classifications		Mechanical Properties									Dimension & Tolerance
		Yield Point N/mm² (min.)		Tensile Strength N/mm²	Yield Ratio % (max.)	Elongation % (min.)			Impact Thickness t > 12		
		Thickness			Thickness	Thickness			Temp	Energy	
		t ≤ 16	16 ≤ t ≤ 40		t ≥ 12	t ≤ 5	5<t≤16	t > 16	°C	J (min.)	
JIS G 3101 :2004	SS400	245	235	400-510	-	21	17	21	-	-	
	SS490	285	275	490-610	-	19	15	19	-	-	
	SS540	400	390	540 min.	-	16	13	17	-	-	
JIS G 3106 :2004	SM400 A	245	235	400-510	-	23	18	22	-	-	JIS 3192 :1990/1994 JIS A 5526 :1988
	SM400 B	245	235	400-510	-	23	18	22	0	27	
	SM490 A	325	315	490-610	-	22	17	21	-	-	
	SM490 B	325	315	490-610	-	22	17	21	0	27	
	SM490 YA	365	355	490-610	-	19	15	19	-	-	
	SM490 YB	365	355	490-610	-	19	15	19	0	27	
	SM520 B	365	355	520-640	-	19	15	19	0	27	
	SM520 C	365	355	520-640	-	19	15	19	0	47	
JIS G 3136 :2005	SN400 A	235	235	400-510	-	17	17	21	-	-	
	SN400 B	235-355	235	400-510	80	18	18	22	0	27	
	SN490 B	325-445	235-355	490-610	80	17	17	21	0	27	
BS 4360 :1990	43A	275	325-445	430-580	-	20			-	-	BS 4 Part 1 :2005
	43B	275	265	430-580	-	20			20	27	
	43C	275	265	430-580	-	20			0	27	
	43D	275	265	430-580	-	20			-20	27	
	43DD	275	265	430-580	-	20			-30	27	
	50A	355	345	490-640	-	18			-	-	
	50B	355	345	490-640	-	18			20	27	
	50C	355	345	490-640	-	18			0	27	
	50D	355	345	490-640	-	18			-20	27	
BS EN 10025 :1993/2004	S275JR	275	265	410-560	-	23			20	27	EN 10034 :1993
	S275JO	275	265	410-560	-	23			0	27	
	S275J2	275	265	410-560	-	23			-20	27	
	S275J2G3	275	265	410-560	-	23			-20	27	
	S355JR	355	345	490-630	-	22			20	27*	
	S355JO	355	345	490-630	-	22			0	27	
	S355J2	355	345	490-630	-	22			-20	27	
	S355J2G3	355	345	470-630	-	22			-20	27	
ASTM :2003	A36	250		400-550	-	20			-		ASTM A6/A 6m :2003
	A 572 Gr.42	290		415 min.	-	20			-	-	
	A 572 Gr.50	345		450 min.	-	18			-	-	
	A 992	345-450		450 min.	85	18			-	-	
DIN 17100	St 33	185	175	290	-	16			-	-	-
	St 37-2	235	225	340-470	-	24			-	-	
	St 44-2	275	265	410-540	-	20			-	-	
	St 50-2	295	285	470-610	-	18			-	-	
	St 52-3	355	345	490-630	-	20			-	-	
AS/NZS 3679.1 :1996		t ≤ 12	12 < t ≤ 40		-				-	-	AS/NZS 3679 :1996
	250	260	250	410 min.	-	22			-	-	
	250L0	260	250	410 min.	-	22			-	-	
	350	360	340	480 min.	-	20			-	-	
	350L0	360	340	480 min.	-	20			-	-	
		t < 11	11≤t≤17	t > 17	-				-	-	
	300	320	300	280	440 min.	-	22		-	-	
	300L0	320	300	280	440 min.	-	22		-	-	
JIS A 5528 :2000	SY295	295			490 min.	-	17		-	-	JIS A 5528 :2000
	SY390	390			540 min.	-	15		-	-	